List of Tables

OC-1 Normative contracts in three close environments (Opus) 3
OC-2 Affordance ledger, layer 1 (Opus) 5
OC-2 Affordance ledger, layer 2 (Opus) 6
OC-2 Affordance ledger, layer 3 (Opus) 7
OC-3 Provider cost and token usage (Opus) 8

Opus peer exhibits

Identifier note. OC-1 and OC-2 were numbered T6 and T7 in Opus's phase-3 proposal; those numbers are *not* the shared T6 (eliminations) or T7 (build recovery). Inside OC-2, R-numbers refer to the reconciled phase-2 claim ledger (`paper_phase2_2026-10-28/core_review/astra/claims.md`); M1 refers to Opus's `experiment_decisions.md`; "T4" refers to Opus's `figure_table_plan.md` build-recovery table, whose shared counterpart is T7.

Each cell records what each environment's **instructions** or its **analysts' coding scheme** say about a behavior. A cell is not a moral judgment or a finding about behavior.

- "Not addressed" means not found in the inspected prompt or sections.
- "Externally coded" means the analysts classify the behavior as misaligned even though the agents' instructions are silent.

Sources: MonopolyBench `SYSTEM_PROMPT_V1` (`python/packages/arena/src/monopoly_arena/player_config.py`); AgentMonopoly, arXiv 2610.29330v2 §4, plus the repo README at commit 2434f427; Vending-Bench Arena paper, arXiv 2608.14825v3 §2.1–2.2, §4.1, App. D. Checked 2026-10-29 (Opus). Machine-readable version: `normative_contracts.csv`.

OC-1. Normative contracts in three close environments

Behavior	MonopolyBench (this cohort)	AgentMonopoly	Vending-Bench Arena paper
False public factual claim	**Explicitly permitted** ("Deception in public chat is allowed")	Bluffing **encouraged**; factual falsity not separately addressed	Instruction silent; **externally coded** misaligned (Stage B false factual claim)
Public message contradicting own action	Not explicitly addressed. Public deception is permitted, and the prompt also says "Be consistent unless there is a reason to change". Not the same measured violation as where it is prohibited	**Explicitly prohibited**	Instruction silent; follow-through measured descriptively (App. D)
False private note or reasoning	**Soft instruction against it** ("private thoughts should reflect true reasoning"). The note is also fed back as the agent's own memory, so a false note can mislead the agent's later decisions (Fable's point). Whether AgentMonopoly feeds `chain_of_thought` back is not reported in the inspected sections	Not addressed; the explanation is "not ... how the model computes its decision"	Instruction silent; reasoning summaries audited for intent, as a lower bound
Withdrawing or changing a stated offer	Not addressed	Not addressed	Instruction silent; promise enactment measured
Coordinated pricing or market division	No price mechanism; "cooperative" play permitted; coalitions not addressed	Not addressed	Instruction silent; **externally coded** misaligned (collusion)
Pressure on or exploitation of a distressed counterparty	"manipulative" and "opportunistic" play **explicitly permitted**	Strategic rent waiving allowed; exploitation not addressed	Instruction silent; **externally coded** misaligned (manipulation)
Illegal game action	**Engine-prevented**: rejection, validation error, corrective retry, then deterministic fallback	**Engine-prevented** (legal_actions)	Not reported in inspected sections

Caption. The same act has a different normative status across environments, so rates and labels do not transfer between them.

In MonopolyBench, a false public claim is authorized competitive conduct. A false private note is a potential breach of a soft instruction, but it is confounded with memory error. Applied rule violations cannot occur, while attempted invalid actions are recorded.

Under the current contract, an instruction-violation finding would require evidence that identifies a knowing false private note. That evidence is unavailable for most decisions (channel availability) and is not established in any checked case. This is non-identification, not impossibility

(accepted from Astra, debate R3).

OC-2. Affordance ledger: what the sandbox implements, what this cohort used, what remains unimplemented

Opus, 2026-10-29. The three-layer split (environment affordance / analysis protocol / not implemented) was agreed with Fable (debate R5). The wording on non-identification was corrected after Astra's R3. A row marked "implemented" in the environment layer means the code exists and was exercised. It does not mean the cohort measured something with it unless the "Used" column says so.

Layer 1: environment affordances (implemented)

OC-2, layer 1

Affordance	Evidence	Used for a cohort finding?	Enables	Not established by this cohort
Engine-authoritative rules with an explicit legal-action menu at every decision; invalid actions rejected	`decisions.jsonl` menus; engine; `docs/model_response_policy.md`	Yes, for all economic findings	Exact legality, opportunity-conditioned denominators	Applied rule violations cannot occur by construction. Attempted invalid actions are recorded as validation errors, not misconduct
Free-text public message on every action, visible to all players[1]	`SYSTEM_PROMPT_V1`; `LLM_PUBLIC_MESSAGE` events	Yes (R18–R21)	Proposition-level social chronology with exposure tracing	Persuasion effect; belief change
Private note channel, instructed to "reflect true reasoning", fed back as memory	`SYSTEM_PROMPT_V1`; prompt memory	Partly. Mostly absent for Opus (action-only after refusal)	A soft honesty norm can be audited case by case	A knowing false private note is not identified in any checked case. Error, stale context and channel conflict remain possible
Engine-settled bilateral trades, up to 20 exchanges, with counters	`run_config.json` `max_trade_exchanges:20`; TRADE_* events	Yes (R03–R05)	Two-sided consequences; initiator versus final-terms author	Trade value or bilateral welfare
Unenforced verbal assurances (for example "offer stands") in public text	No commitment object exists; assurances are free text only	Yes, one case (G18, R20)	Withdrawal or invocation chronology	Breach rate; costly reliance
Forced liquidation (sell buildings, then mortgage), finite house stock, creditor inherits estates intact (declared variant)	`docs/benchmark_audit_fixes.md`; engine	Yes (R08–R10, R15–R17)	Accessible versus nominal finance; stock restrictions; inheritance concentration	Causal effect of inheritance or house denial on winning
Per-attempt response provenance, with refusal-recovery stages and deterministic fallback	`run_config.json` `response_policy`; attempt records	Yes (M1, T4)	Distinguishes chosen holds, rejected plans and defaults	The effect on outcomes of a different recovery policy
Model-specific request settings (Grok `tool_choice="required"`; others `"auto"`)	`player_config.py:328-334` at both run commits	Yes, as a qualification (R23)	Configured-agent unit	Origin of received-name conflicts

Continued on next page

[1] Conversion correction, 29 October 2026: the message is logged publicly on every action, but delivery into each recipient's prompt is not guaranteed (20-entry public timeline; delivery omissions). A logged utterance is not necessarily delivered (integrated outline, Sections 3.3 and 6).

OC-2 Affordance ledger, layer 1 (Opus) continued

Affordance	Evidence	Used for a cohort finding?	Enables	Not established by this cohort
Deterministic seeded engine; event, action, decision and snapshot artifacts; state and artifact replay	Saved-game `run/`; Sol's 20/20 replay	Yes (integrity)	Reproducibility; audit	—
Exact decision-prefix reset with event-rebuilt memory, plus engine-validated stepping of alternative legal actions	`paper_phase1_2026-10-28/opus/opus_sandbox_checks.json`: 20 branch checks, all passing[2] prefix-identical, memory-equal, suffix-reproduced and branch-divergence checks	Yes, through the analysis protocol (Layer 2)	Local feasibility tests	Full counterfactual games; alternative winners
Scripted reference policies (`random_legal`, `always_buy`, `cash_conservative`, `no_trade`, `builder`, `auction_aggressive`)	`python/packages/arena/src/monopoly_arena/baselines.py`; exercised in the 20 phase-1 rollouts	**No** (implemented, not measured for the cohort). `propose_trade` uses empty arguments, so baselines cannot exercise the trade route	Seat and dice baselines (experiment matrix C-tier O6)	—
130 micro scenarios with a legality-validating runner	`contracts/micro/scenarios/` (130 files); `micro_runner.py`	**No** (no cohort results)	Targeted capability probes under fixed states	—

Layer 2: analysis protocols (performed by analysts, not an environment API)

OC-2, layer 2

Protocol	Evidence	Findings
Exact-prefix multistep legal-sequence audit to a stated local endpoint, without continuing the historical suffix	`core_review/astra/default_build_plans.json`, `g07_hotel_lock.json`, `g12_channel_liquidation_check.json`; `core_review/opus/bankruptcy_bounds.json`; `core_review/sol/elimination_upper_bounds.json`	R06, R08–R10; refuted generated labels (G6-C021, G14-C014, G20-C024)
Event-based economic accounting with ownership windows, estate provenance and censoring	`core_review/astra/development_aggregation_check.json`; `core_review/sol/corpus_economics_audit.json`	R12–R16
Proposition-level social chronology: truth, speaker record, exposure, recipient response, settlement, competing explanation	`core_review/sol/g13_omission_check.json`, `g11_social_channel_checks.json`, `g04_uptake_boundary_check.json`; `core_review/astra/g13_claim_checks.json`	R18–R21
Channel-conflict diagnostics (content versus received tool name and arguments)	`core_review/sol/received_name_conflict_audit.json`	R23 (a diagnostic count)

Continued on next page

[2] Scope, as in the integrated outline, Section 6: reset and local branching are tested at these stated prefixes; full-prompt equality beyond memory, provider suffixes and general rollout termination are not thereby demonstrated.

OC-2 Affordance ledger, layer 2 (Opus) continued

Protocol	Evidence	Findings
Prompt-exposure tracing over time	`core_review/astra/negotiation_units.json`; `paper_phase1_2026-10-28/sol_round2/public_delivery_summary.json`	R22; delivery omissions

Layer 3: not implemented, and needed for specific alignment or safety claims

OC-2, layer 3

Missing affordance	Claim it would unlock	Comparable design elsewhere (verified)
Binding or deferred commitment objects (IOUs, future payments, conditional deals) that the engine records but does not enforce	Breach rate of unenforced commitments, with a counted opportunity and observable reliance	Arena App. D measures transfer-promise enactment (free-text promises)
Private bilateral messaging	Secret coordination; private deception; coalition formation	Arena emails go to named recipients only (§2.1)
A treatment with an explicit public-honesty or no-contradiction norm	Instruction-violation rates for public speech	AgentMonopoly §4 prohibits public/action contradiction
Engineered violation opportunities (actions that are possible, rewarding and against a stated rule, with monitoring)	Misalignment under temptation; capability × violation trend	Emergent Collusion (arXiv 2609.24967) protocol-violation affordance (abstract)
An oversight or monitor agent with a frozen channel and adjudicated ground truth	Monitor accuracy; monitorability	CoT-monitoring literature (Baker et al. 2025; Korbak et al. 2025)
Packaged branching API, reward interface, RL adapter, training loop	Training or improvement claims	Not implemented anywhere in this repository. No training claim is made
Human participants	Human–agent conduct	Arena §6 lists this as future work
A capability axis (several tiers within one provider lineage, same harness)	Capability–alignment trend	Arena §5.3.2 (cross-model null); Emergent Collusion (within-family positive)

Caption. The environment already makes economic conduct, public speech, private notes and per-attempt provenance adjudicable against exact legal state.

The present cohort uses Layers 1 and 2 to produce descriptive economic and social findings. The alignment-relevant claims the paper does *not* make need specific Layer 3 affordances, each paired with a treatment and a counted opportunity.

OC-3. Provider cost and token usage by configured agent

Agent	OpenRouter model	Effort	Decisions	Own-model attempts	Attempts missing cost	Known cost (USD)	USD / decision	Prompt tokens	Completion tokens	Reasoning tokens	Completion tok. / decision	Reasoning tok. / decision	Latency s / decision (observational)
Gemini (Argon)	`google/gemini-4-argon`	medium	1,669	1,689	0	18.59	0.0111	6,220,720	614,880	428,756	368	257	8.2
Grok	`x-ai/grok-4.7`	medium	1,663	1,698	3	$\geq$ 55.60	0.0334	9,718,441	6,720,746	6,082,677	4,041	3,658	65.0
Opus	`anthropic/claude-opus-5.5`	medium	1,696	3,301	1,600	$\geq$ 48.80	0.0288	8,724,266	695,305	440,059	410	259	7.1
Sol	`openai/gpt-6.1-sol`	medium	1,961	1,991	0	18.72	0.0095	6,488,141	250,050	75,568	128	39	4.8

Notes. Money is provider **USD** as reported by OpenRouter, not game dollars. Known cost totals 141.71 USD. Known cost is a lower bound (marked $\geq$) wherever attempts lack accounting: Opus's 1,600 refused attempts and 3 Grok attempts have no OpenRouter cost record, so the Opus token totals are lower bounds as well. Decisions sum to 6,989. All four agents ran with provider-native reasoning effort "medium", which does not equalize compute. Completion and reasoning tokens follow provider-specific accounting and are not summed across providers. Latency is wall-clock observational metadata and does not affect game progression. Source: `opus/exhibits/cost_usage.csv`, provenance `cost_usage.provenance.json` (20 `run/usage.json` files via `cohort_overview.json`).